

Imaging of Climbing Injuries

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Jens Liße in „Stalingrad direkt“, Frankenjura, Germany,
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Radiological imaging is essential for correct diagnosis of climbing injuries. Classical X-ray procedures and computed tomography (CT) as cross-sectional imaging are excellent for detecting bony changes, such as degeneration, fracture, or avulsion. However, they have limited soft tissue contrast and are associated with exposure of the patient to ionizing radiation. Sonography is widely applied, is easy to learn, and can be performed functionally during the patient's movement. It is suited for the primary diagnosis of soft tissue injuries caused by climbing, such as the pulley rupture. Magnetic resonance imaging (MRI) offers better soft tissue contrast than any other cross-sectional imaging and is established for further diagnosis of numerous musculoskeletal issues in climbing.

5.1 X-ray

5.1.1 X-ray Procedures for the Diagnosis of Climbing Injuries

The X-ray procedures used in diagnostics today include classical projection radiography (synonymous with “conventional X-ray” or “classic X-ray”) and other procedures derived from it, such as fluoroscopy or computed tomography (CT). All procedures are suited for the diagnosis of bone pathologies but with limited soft tissue contrast compared to sonography or MRI. Regarding climbing, X-ray and CT are particularly important; fluoroscopy is less so.

In all acute trauma, e.g., a fall or a major distortion, the primary goal is to detect an acute bony pathology (i.e., fracture). In these cases, conventional X-ray is the primary imaging method of choice. However, some anatomical regions may be difficult to assess with X-ray because of the superimposition of many small or complex bony structures, for example, the wrist or the tarsus. In these regions, additional X-ray projections with special settings (e.g., so-called scaphoid targeting at the wrist, Figs. 5.12, 5.13, 5.14, and 5.15) may be useful. When X-ray diag-

nosis remains unclear and the patient is still suspicious for fracture, additional cross-sectional imaging should be performed. CT is primarily indicated, allowing exact fracture diagnosis in more difficult anatomical regions.

In patients without acute trauma and with gradual developing complaints, X-ray and CT are less important, as these issues are mostly caused by soft tissue injuries. In most cases, a fracture may already be excluded clinically. Nevertheless, X-ray is part of the basic examination. The primary aim is to rule out chronic bone pathology, such as degeneration, osteophyte formation, and soft tissue calcification, but also bone tumors, bony cysts/ganglia, or systemic changes.

One major exception are stress fractures, which may result from chronic repetitive microtrauma, such as in the hamate hook [1, 2] or the middle finger growth plate of teenagers [3]. Stress fractures of the hand are a unique characteristic in climbing and are best detected with MRI. Trabecular microfracturing is usually too small to be visible with X-rays or CT but can be detected due to an accompanying bone marrow edema with improved soft tissue contrast during MRI.

5.1.2 Tips for X-ray Procedures

X-ray examinations should be performed by trained technicians. Standard images are defined for almost every joint and every region of the body and are a basic requirement for reliable imaging. Long-term expertise is recommended to learn proper radiological assessment. An open approach to X-ray imaging in the everyday clinical routine is useful. We encourage a regular discussion of findings with partners in radiology, orthopedy, and trauma surgery. X-ray image quality should be adequate, and standards of adjustment techniques are necessary for generating proper imaging. Typically, X-ray in two planes consists of a correct projection in an anterior-posterior and lateral plane with clear identification of joint surfaces. Anatomically complex regions, such as the wrist, require special imaging, such as the scaphoid target image or

the ball image. If the bone findings are unclear, a supplementary CT is usually required.

5.1.3 Fundamentals of X-rays

All X-ray procedures rely on weakening of radiation in the human body. This is greater in thicker and denser objects and increases with the charge of atomic nucleus of tissue. With different degrees of attenuation, an image of the inside of the body can be taken, and thus, for example, bones can be distinguished from soft tissue. In conventional X-rays, three-dimensional objects are summed up in the direction of the projection plane. Thus, the result of X-ray imaging is two-dimensional. In consequence, injuries are often not accessible in only one projection plane. A dislocated fracture, for example, may not be distinguished in anterior-posterior projection, whereas this would be visible in lateral projection.

CT is also based on X-ray weakening, with the difference that an X-ray tube rotates around the human body during imaging, thus producing cross-sectional images without any overlap. Based on the acquired radiation attenuation per angle of rotation, density values can be assigned to each volume unit (voxel) by a computer calculation. These density values can be displayed in grayscale images along any slice plane without overlapping, including 3D techniques (e.g., with so-called volume rendering views, Fig. 5.18; maximum intensity projections, Fig. 5.17; or multiplanar reconstructions, Fig. 5.16). Due to the exposure of the patient to ionizing radiation, the operation of all X-ray-based systems is usually subject to strict safety regulations.

5.2 Sonography

5.2.1 Sonography for the Diagnosis of Climbing Injuries

Sonography (ultrasound or “US”) is the standard first-line imaging modality for diagnosing

climbing-associated soft tissue injuries. The major advantage is that it can be performed functionally, i.e., during movement or when the muscles and tendons are stressed. It is particularly suitable for diagnosing hand and finger pathologies due to the low tissue penetration depth required. At the same time, it enables high detail resolution. Regarding climbing injuries, US is also important for elbow and shoulder diagnostics. In principle, it can be applied to all joints and regions of the human body.

Functional US examination has become a well-established standard in the detection of pulley ruptures [4]. Bowstringing sign under forced finger flexion (Figs. 5.1 and 5.2) is a strong indicator for pulley rupture. By measuring the distance between bone and tendon, this sign allows definition of the severity and location of a pulley rupture. In trigger finger inflammation (tenosynovitis stenosans), increased friction or chronic inflammatory thickening of the pulleys and tendons may be imaged functionally during movement. Functional examination may be designed

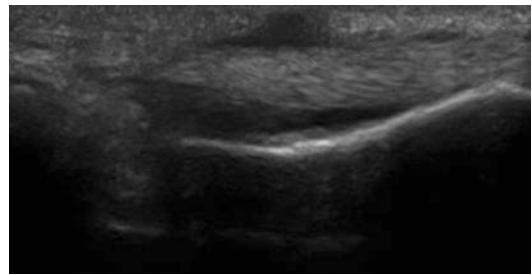


Fig. 5.1 Sonography under provocation, massive bowstringing over the A2 pulley

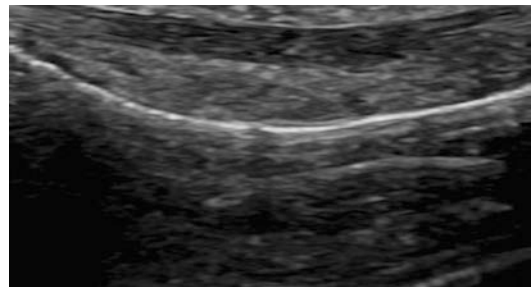


Fig. 5.2 Sonography under provocation, massive bowstringing over the A4 pulley

individually, dependent on the joint and the patient clinics. For shoulder rotator cuff injuries, a wide range of functional sonographic examination techniques has been described.

Fluid collections, tenosynovitis, and ganglion cysts can be identified by the characteristics of echo-free (black) water with dorsal sound amplification (white in the image). Bones and calcifications can be identified by total reflection. This means that the surface of the corresponding structure appears white, whereas the zone behind the object appears black. Ultrasound allows the detection of muscular injuries, such as fiber tear, bleeding, or discontinuity with definition of the resulting severity. The intramuscular viscoelastic tissue characteristic may be examined with the so-called elastography. Small image shifts during pressure application with the transducer probe are evaluated in a computer analysis. As a result, the stretching ability of the muscles can be displayed in spatial resolute image. A particularly well-established sonographic application is so-called duplex sonography, in which the blood flow within vessels or tissue is displayed in color-coded images using the Doppler effect.

5.2.2 Tips for Sonography

Ultrasound devices may be compact and easily portable resembling laptop design. This allows acute trauma examination, for example, during a competition. We recommend modern ultrasound devices in combination with high-resolution linear transducer probes. High megahertz transducers (typically 10–20 MHz) are particularly suitable for hand and finger examinations. Sonography depends on the individual expertise of the examiner; thus, proper training is helpful. Finger examinations benefit from short linear transducers with a small contact surface, as this hardly impairs the patient's freedom of movement during functional examinations. Mechanical sound wave coupling is crucial for good results, and air between the patient and the transducer probe must be avoided. In order to ensure sound coupling, ultrasound gel can be used, whereas functional hand and finger examinations can also

be performed elegantly in a water bath. We recommend a standardized setup, in which the patient and examiner sit opposite each other with the patient's hand on a stable examination table.

5.2.3 Fundamentals of Sonography

Ultrasonic waves are mechanical vibrations that propagate in the human body in a longitudinal direction in the form of compression and decompression. When passing through tissue, the sound waves are weakened by various effects, such as absorption, refraction, scattering, divergence, and reflection. The latter effect is particularly noticeable at tissue interfaces with different acoustic properties. The sound waves are particularly strongly reflected at locations with so-called high acoustic impedance. The reflected sound waves are converted into electrical signals in a specially arranged system of piezoelectric elements in the transducer probe. Based on these signals, the location-coded image is reproduced by a computer in real time as a high-contrast image corresponding to the acoustic tissue properties. The power of ultrasound waves increases with increasing frequency, while, at the same time, the penetration depth decreases.

5.3 Magnetic Resonance Imaging (MRI)

5.3.1 MRI for the Diagnosis of Climbing Injuries

Magnetic resonance imaging is particularly important for further diagnostics of climbing injuries. It has a better potential to depict pathological and post-traumatic soft tissue conditions with higher spatial resolution and contrast than any other imaging modality. The slice direction can be chosen freely in any plane and/or in three dimensions. Numerous special MRI techniques are available, including application of intravenously or intra-articular contrast agent. MRI is suitable for all joints and regions of the body. Dedicated examination protocols with specially

designed anatomical surface-receiving coils are available for almost every joint.

The MRI detectability of all climber-specific injuries of the hand and fingers is excellent. Among these are all flexor tendon injuries, extensor tendon injuries, tenosynovitis, capsular lesions, volar plate injuries, lumbrical muscle injuries, Dupuytren's disease, bony pathologies such as epiphyseal joint injuries, and many more.

Various MRI techniques for further diagnostics of the pulley system have been described. Pulley ruptures can be identified indirectly by bowstringing with the same approach as in sonography. In addition, however, the morphology and localization of the individual ruptures can be displayed directly. Direct pulley injury imaging requires particularly powerful MR scanners with strong magnetic field strength, enabling high spatial resolution [5, 6]. It allows definition of the exact location and shape of the rupture, which is advantageous for presurgical evaluation. Complications, such as a dislocation of a ruptured pulley ligament between flexor tendons and bone, are easy to detect. This has been described in literature as the so-called FLIP phenomenon (flap irritation phenomenon, Fig. 5.4).

The standard MRI is performed in a neutral position. However, functional examination is possible, requiring additional time (Fig. 5.5). For example, a hand may also be examined in the crimp-grip position [7]. Dynamic examinations of the fingers may be performed with MR cinematography [8]. This allows evaluation of further indirect physiological and pathological movement patterns of the tendons and the joints. This may be particularly useful regarding the volar plate and the A3 pulley. Despite not being standard, functional examinations may also contribute to better diagnostics in other joints, such as the shoulder for diagnosing instability (e.g., ABER positioning; abduction elevation external rotation).

MR arthrography (Fig. 5.21) requires administration of intra-articular contrast agent and enables a better assessment of injuries to the capsular ligaments. This technique is regularly performed for climbers. It is particularly useful for detecting injuries to the biceps tendon anchor and

the glenoid labrum in the context of SLAP injury (superior labrum anterior to posterior). In the hip joint, it improves diagnostics of an femoroacetabular impingement (FAI).

Many other modern MR techniques are available, allowing a more profound and specific joint diagnostic. One example is cartilage imaging, in which the quality of cartilage is assessed compositionally, using mapping techniques with analysis of the cartilage quality. The wide range of all possible modern MR applications would go beyond the scope of this chapter.

5.3.2 Tips for MRI

A basic understanding of the different MRI sequences is useful. The specific image contrast depends on the sequence weighting (see Sect. 4.2.3). In T2-weighted sequences, fat and water are particularly rich in signal (hyperintense, bright on the image), whereas in T1-weighted sequences, only fat is rich in signal (note: T2, fat and water; T1 only, fat). Other tissues, such as tendons, capsules, connective tissue, and compact bone, are usually low in signal (hypointense, dark). A precise diagnosis of these structures requires improved techniques, such as proton density (PD)-weighted sequences. In PD imaging, both fat and water are rich in signal, but the detail recognition of the previously mentioned low-signal connective tissue structures is better. For musculoskeletal MRI and thus for diagnostics of climbing injuries, PD sequences should always be included in the examination protocol.

One common characteristic of all of these MRI sequences is a relatively bright image, representing hyperintense fat and water. In contrast, sequences can also appear almost completely dark. In these images, a fat saturation (fs) technique was probably applied. Fat saturation (fs) techniques can be combined with the previously mentioned bright standard sequences (corresponding sequence names, e.g., T1 fs, T2 fs, PD fs), which will increase the acquisition time slightly. The main advantage is the higher sensitivity for injuries as a bright area in the overall dark image. With some practice, even an inexpe-

rienced MRI viewer may be able to diagnose relevant musculoskeletal injuries on PD fs sequence (Fig. 5.8 and 5.9).

It is beneficial to include approximately the same number of sequences with and without fs in an MRI. The choice of correct slice plane (e.g., axial, coronal, sagittal, or 3D image) and slice thickness according to the expected injury is of utmost importance. The slice thickness should match the size of the joint. For example, a slice thickness of 2 mm is recommended for finger and wrist examinations, whereas a slice thickness of 3–4 mm is usually enough for larger joints, such as the hip or knee. As sport-specific injuries often result in subtle joint changes, application of modern MRI scanners with dedicated multichannel surface coils in combination with high magnetic field strength is recommended whenever possible. For instance, a 3 Tesla scanner is potentially superior to a 1.5 Tesla scanner, resulting in better image quality.

5.3.3 Basics of MRI

MRI is the result of nuclear physical effects in the permanently strong magnetic field of a superconducting magnet. Outside the MRI scanner, the spins of hydrogen atoms in the human body are mostly disordered. Under the influence of the strong magnetic field in the MRI, the nuclear spins are aligned, resulting in a measurable magnetization along the main magnetic field axis. The nuclear spins perform a directed “gyroscopic motion” along this axis; this is called “precession” in physics. The precession frequency increases proportionally with the magnetic field strength; it is 42.48 MHz/Tesla. For imaging, artificial attenuations of the magnetic field strength are performed with gradient coils in different spatial directions, which allows an exact location encoding of the MR signals. In addition, high-frequency (HF) pulses are transmitted with transmitter coils under resonance conditions. High-frequency pulses lead to a targeted “excitation” of the nuclear spins, resulting in longitudinal and transverse magnetization. After termination of an RF pulse, spins lose the

absorbed energy by emitting the MR signal; this “relaxation process” can be detected with receiving coils.

MRI imaging is performed by repeated application of HF pulses with intermediate detection of MR signals. Such a sequence of excitation and relaxation of the spins is called an MR sequence. A spatial assignment of the received MR signal to a corresponding coordinate in the later slice image is done in complex computer calculations. Due to the different distribution of water and fat in the body and the different relaxation characteristics of different tissues, a particularly high-contrast image of soft tissues can be obtained with correspondingly differently designed MR sequences.

5.4 Typical Differential Diagnosis of Climbing Injuries

5.4.1 Pulley Rupture

A 23-year-old climber noticed a popping sound with acute severe pain in his right ring finger while climbing. This is a severe injury with the rupture of the A2, A3, and A4 ring ligaments (i.e., triple pulley rupture). The patient received a surgical pulley reconstruction due to the massive functional loss with risk of subsequent flexion contracture and the evidence of a flap irritation phenomenon (FLIP) (Figs. 5.1, 5.2, 5.3, 5.4, and 5.5).

5.4.2 Stress Fracture of the Growth Plate

A 14-year-old ambitious climbing athlete presented with pain and swelling over the PIP joint of the left middle finger, which gradually appeared over the course of several weeks. The diagnosis was a typical finding of an epiphyseal stress fracture of the middle phalanx base. A training stop for 3 months and cast splinting was performed. Since X-ray diagnostics could not show the stress fracture in this case, a follow-up



Fig. 5.3 Sagittal PD fs: triple rupture of the ligament with hematoma in the tendon sheath and bowstringing over the A2 and A4 pulleys. In the neutral position, no clear bowstring over the A3 pulley

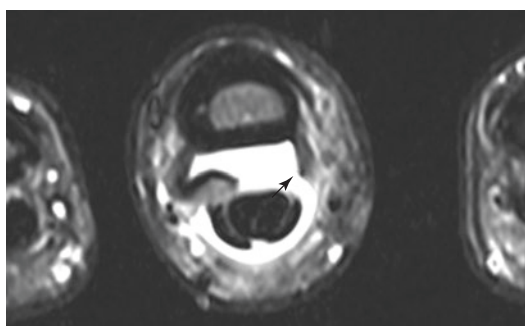


Fig. 5.4 Axial PD fs: A2 pulley rupture with dislocation of the ligamentous stump between the flexor tendon and phalanx in the sense of a flap irritation phenomenon (FLIP)

MRI was performed to confirm fracture healing after 3 months (Figs. 5.6, 5.7, 5.8, and 5.9).

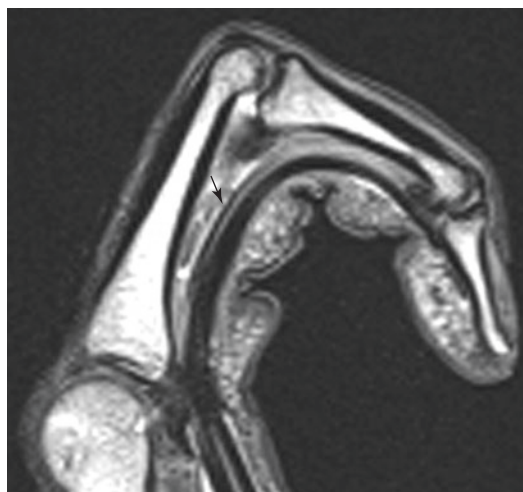


Fig. 5.5 Maximum finger flexion achievable by the patient in the MR functional examination with sagittal PD: triple (A2-A3-A4) pulley rupture with increased bowstringing over the phalanges and also over the A3 pulley (note the increased distance of the flexor tendons to the volar plate of the PIP joint)

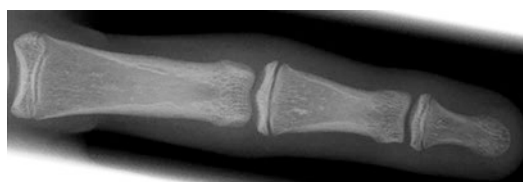


Fig. 5.6 X-ray in a.p. projection: soft tissue swelling over the PIP joint without clear evidence of fracture



Fig. 5.7 Lateral X-ray projection: soft tissue swelling over the PIP joint without clear evidence of fracture

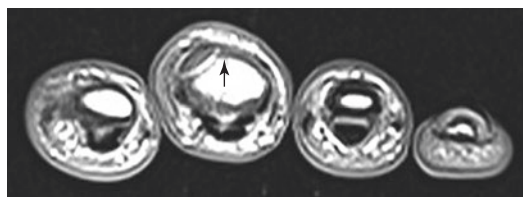


Fig. 5.8 Axial T1-weighted MRI: hyperintense fracture line in the middle phalanx base Dig. 3

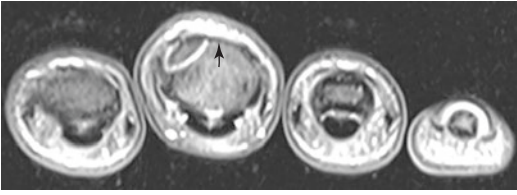


Fig. 5.9 Axial PD weighted fs MRI: hyperintense fracture line in the middle phalanx base Dig. 3, the fracture is easier to detect compared to Fig. 5.8 due to fat suppression (fs)

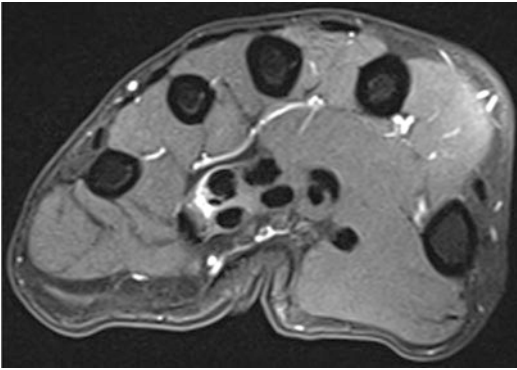


Fig. 5.10 Axial PDfs MRI: hyperintense hematoma surrounding the fourth flexor digitorum profundus tendon with tear of the lumbrical muscle tendon at the origin in the sense of a grade III lumbrical shift syndrome

5.4.3 Lumbrical Shift Syndrome

A 37-year-old female climber presented with acute pain and swelling in the palm of her hand on the right side above the flexor tendon of the ring finger. The injury healed conservatively within 2 months without functional sequel (Figs. 5.10 and 5.11).

5.4.4 Trabecular Microfracture of the Scaphoid

A 16-year-old climber fell on his wrist while bouldering. Unlike MRI, the scaphoid series and CT did not show this microfracture. The injury healed conservatively after 8 weeks in a cast (Figs. 5.12, 5.13, 5.14, 5.15, 5.16, 5.17, 5.18, and 5.19).



Fig. 5.11 Coronary PDfs MRI: hyperintense hematoma along the flexor digitorum profundus tendon Dig. 4 and the third lumbrical muscle

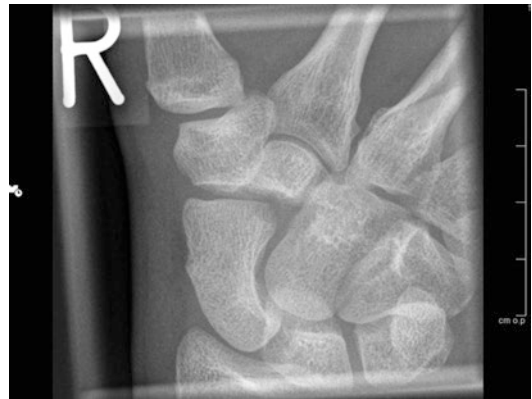


Fig. 5.12 Inconspicuous scaphoid series

5.4.5 MR Arthrography for Diagnosing Labrum Tear

A 33-year-old climber presented with chronic shoulder problems, pain during the night at rest, recurrent subluxation events, and a feeling of



Fig. 5.13 Inconspicuous scaphoid series



Fig. 5.15 Inconspicuous scaphoid series



Fig. 5.14 Inconspicuous scaphoid series

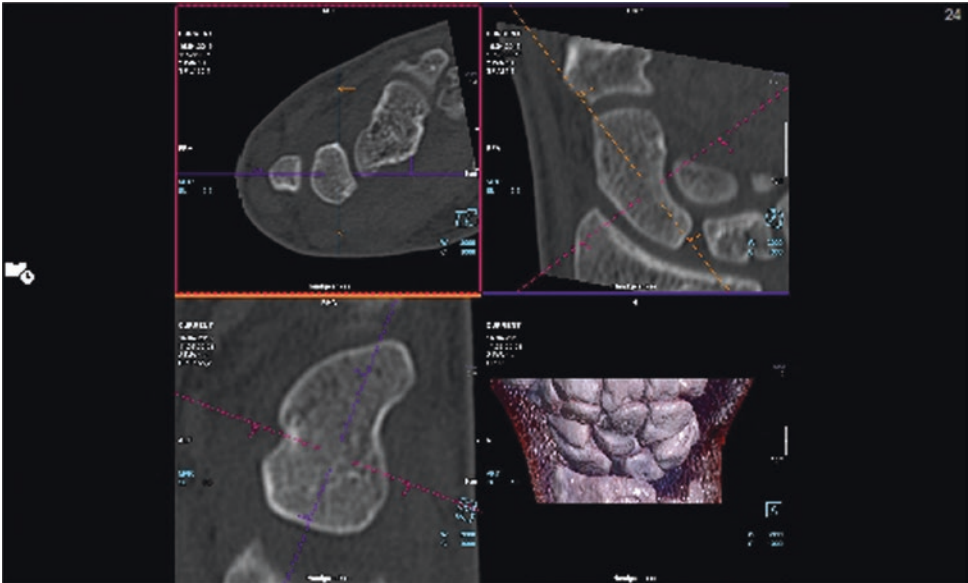


Fig. 5.16 Inconspicuous computed tomography: multiplanar reconstruction (MPR)

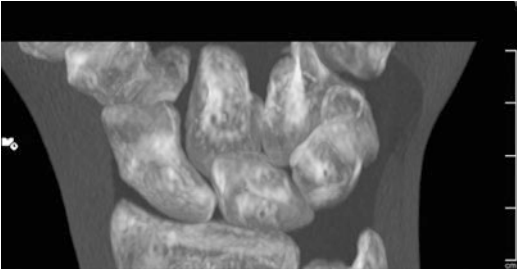


Fig. 5.17 Inconspicuous computed tomography: maximum intensity projection (MIP)

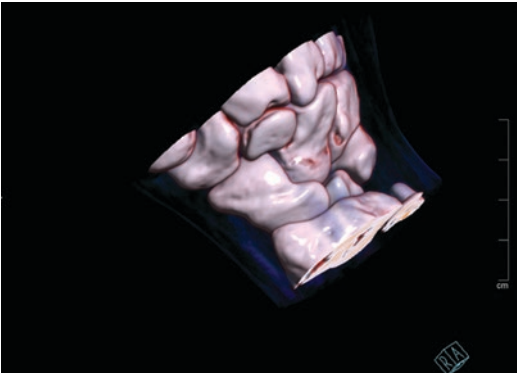


Fig. 5.18 Inconspicuous computed tomography: volume rendering

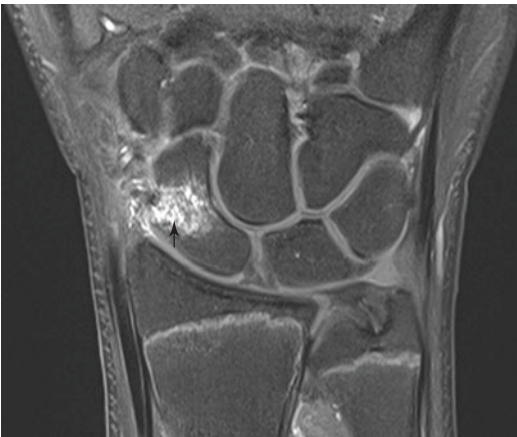


Fig. 5.19 Axial PDfs MRI: detection of a trabecular microfracture of the scaphoid with hyperintense blurred fracture zone

instability while climbing. The suspected labrum tear could be visualized particularly

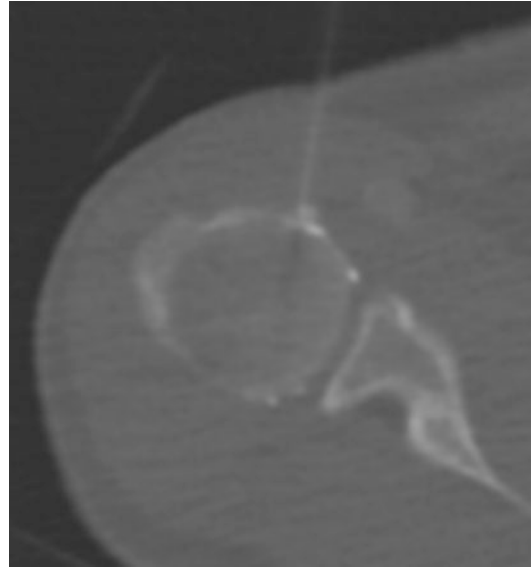


Fig. 5.20 CT-guided puncture of the right shoulder joint for application of MRI contrast agent

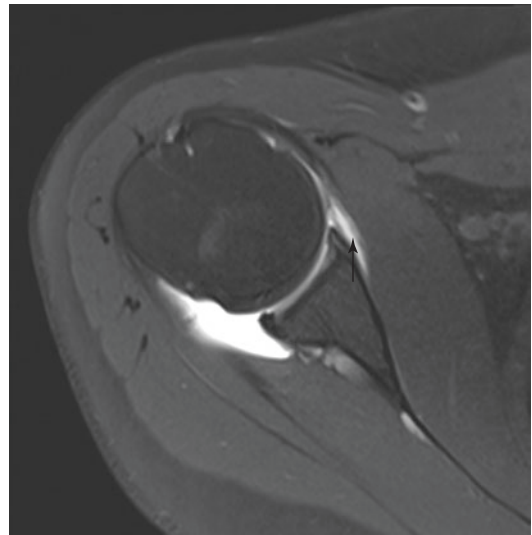


Fig. 5.21 MR arthrography with evidence of a tear in the ventral labrum without lifting of the periosteum, a so-called Perthes lesion

well in MR arthrography, where contrast agent had been applied intra-articularly previously. A surgical labrum repair was performed (Figs. 5.20 and 5.21).

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